Cloud Orchestration APIs

NetApp Keystone

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Cloud Orchestration APIs

The Cloud Orchestration APIs NetApp Service Engine support connecting Cloud Volumes Services to NetApp Service Engine and to work with them from within the NetApp Service Engine. Azure NetApp Files (ANF) and Google Cloud Platform (GCP) are supported for Cloud Volumes Services.

Subscriptions to Cloud Volumes Services are managed outside of NetApp Service Engine. The relevant credentials are provided to NetApp Service Engine to allow connection to the cloud services.

This section describes the APIs for:

- Managing Cloud Volumes Service instances
- Managing Azure NetApp Files accounts
- Managing Azure NetApp Files capacity pools
- Managing Azure NetApp Files Volumes
- Managing Google Cloud Volumes

Cloud Volumes Service instances

Use the methods listed in the following table to retrieve and create Cloud Volumes Service instances.

HTTP Method	Path	Description
`GET`	`/v2.1/cvs/instances`	Retrieve all Cloud Volumes Service instances.
`POST`	`/v2.1/cvs/instances`	Create new Cloud Volumes Service instances.
`GET`	`/v2.1/cvs/instances/{id}`	Retrieve a Cloud Volumes Service instance by ID.
`GET`	`/v2.1/anf/accounts`	Retrieve Azure NetApp Files accounts.
`POST`	`/v2.1/anf/accounts`	Create an Azure NetApp Files accounts
`GET`	`/v2.1/anf/accounts/{id}`	Retrieve an Azure NetApp Files account by ID.
`GET`	`/v2.1/anf/pools`	Retrieve all capacity pools.
`POST`	`/v2.1/anf/pools`	Create a capacity pool.
`GET`	`/v2.1/anf/pools/{id}`	Retrieve an Azure NetApp Files capacity pool by ID.

Cloud Volumes Service instance attributes

The following table lists the Cloud Volumes Service instance attributes.

Attribute	Type	Description
`id`	String	Unique identifier for the Cloud Volumes Service instance.
`name`	String	Name of the Cloud Volumes Service instance.
`cc_working_env`	String	Working environment name within Cloud Manager.
`cc_client_id`	String	Auth0 service ID.
`cc_refresh_token`	String	Auth0 refresh token.
`cc_account_id`	String	Cloud Central account ID.
`cc_account_name`	String	Cloud Central account name.
`cc_credentials_id`	String	ID of the Cloud Centra working environment.
`azure_client_id`	String	Application (Client) ID. The ID of an Active Directory service principal that Cloud Manager can use to authenticate with Azure Active Directory.
`azure_client_secret`	String	The value of client secret for service principal application.
`azure_tenant_id`	String	The Azure Active Directory ID. This is also known as tenant ID.
`azure_subscription_id`	String	The Azure Active Directory subscription ID.
`subtenant_id`	String	The subtenant ID.
`tags`		The key-value pairs.

Retrieve Cloud Volumes Service instance

HTTP Method	Path	Description	Path Parameters
`GET`	`/v2.1/cvs/instances`	Retrieve Cloud Volumes Service instances.	`tenant_id:` (Optional) Return the Cloud Volumes Service instances belonging to the specified tenant. See also Common Pagination parameters.

Required request body parameters: `none`

Retrieve Cloud Volumes Service instance by ID

Use the method listed in the following table to retrieve a Cloud Volumes Service instance by its identifier.

HTTP Method	Path	Description	Parameters
`GET`	`/v2.1/cvs/instances/{id}`	Retrieve a Cloud Volumes Service instance by ID.	`id (string):` The unique identifier of the Cloud Volumes Service instance.

Required request body attributes: `none`

Create Cloud Volumes Service instances

Use the method listed in the following table to create a new Cloud Volumes Service instance.

HTTP Method	Path	Description	Parameters
`POST`	`/v2.1/cvs/instances`	Create a Cloud Volumes Service instance.	None

Required request body attributes: `name`, `cc_working_env`, `cc_client_id`, `cc_refresh_token`, `cc_account_id`, `cc_account_name`, `azure_client_id`, `azure_client_secret`, `azure_tenant_id`, `azure_subscription_id`, `subtenant_id`

Request body example:

```
{
  "name": "instance1",
  "cc_working_env": "my-working-env",
  "cc_client_id": "Mu0V1ywgYteI6w1MbD15fKfVIUrNXGWC",
  "cc_refresh_token": "y1tMw3lNzE8JL9jtiE29oSRxOAzYu0cdnwS_2XhjQBr9G",
  "cc_account_id": "account-335jdf32",
  "cc_account_name": "my-account-name",
  "cc_credentials_id": "d336c449-aeb8-4bb3-af28-5b886c40dd00",
  "azure_client_id": "53ba6f2b-6d52-4f5c-8ae0-7adc20808854",
  "azure_client_secret": "NMubGVcDqkwwGnCs6fa01tqlkTisfUd4pBBYgcxxx=",
  "azure_tenant_id": "53ba6f2b-6d52-4f5c-8ae0-7adc20808854",
  "azure_subscription_id": "1933a261-d141-4c68-9d6c-13b607790910",
  "subtenant_id": "5d2fb0fb4f47df00015274e3",
  "tags": {
    "key1": "Value 1",
    "key2": "Value 2",
    "key3": "Value 3",
    "keyN": "Value N"
  }
}
```

Manage tags for Cloud Volumes Service instances

Use the method listed in the following table to specify tags for the named Cloud Volumes Service instance.

HTTP Method	Path	Description	Parameters
`POST`	`/v2.1/cvs/instances/{id}/tags`	Manage tags for a Cloud Volumes Service instance.	`id (string)`: The unique identifier of the Cloud Volumes Service instance.

Required request body attributes: `key-value pairs`

Request body example:

```
{
  "env": "test"
}
```

Azure NetApp Files accounts

Azure NetApp Files accounts attributes

The following table lists the Azure NetApp Files account attributes.

Attribute	Type	Description
`id`	String	The unique identifier for the Azure NetApp Files account.
`name`	String	The name of the Azure NetApp Files account.
`resource_group`	String	The Azure resource group.
`location`	String	The Azure location (region/zone).
`cvs_instance_id`	String	The Cloud Volumes Service instance identifier.
`tags`	–	The key-value pairs.

Retrieve Azure NetApp Files accounts

HTTP Method	Path	Description	Path Parameters
`GET`	`/v2.1/anf/accounts`	Retrieve Azure NetApp Files accounts.	`subtenant_id:` (Mandatory) The subtenant ID to which the Azure NetApp Files account belongs. `tenant_id:` (Optional) Returns the Azure NetApp Files accounts belonging to the specified tenant. See also Common Pagination parameters.

Required request body parameters: `none`

Retrieve Azure NetApp Files account by name

Use the method listed in the following table to retrieve an Azure NetApp Files account by name.

HTTP Method	Path	Description	Parameters
`GET`	`/v2.1/anf/accounts/{name}`	Retrieve an Azure NetApp Files account by name.	`name (string):` (Mandatory) The name of the Azure NetApp Files account. `subtenant_id (string):` (Mandatory) The subtenant ID to which the Azure NetApp Files account belongs.

Required request body attributes: `none`

Create Azure NetApp Files accounts

Use the method listed in the following table to create a new Azure NetApp Files account.

HTTP Method	Path	Description	Parameters
`POST`	`/v2.1/anf/accounts`	Create a new Azure NetApp Files account.	None

Required request body attributes: `name, resource_group, location, cvs_instance_id`

Request body example:

```
{
  "name": "string",
  "resource_group": "string",
  "location": "string",
  "cvs_instance_id": "5d2fb0fb4f47df00015274e3",
  "tags": {
    "key1": "Value 1",
    "key2": "Value 2",
    "key3": "Value 3",
    "keyN": "Value N"
  }
}
```

Azure NetApp Files capacity pools

Capacity pools attributes

The following table lists the capacity pool attributes.

Attribute	Type	Description
`id`	String	The unique identifier for the capacity pool.
`name`	String	The name of the capacity pool.
`resource_group`	String	The Azure resource group.
`location`	String	The Azure location (region/zone).
`size`	Integer	The size of the capacity pool in TB.
`service_level`	String	The service level name applicable: Ultra, Premium, or Standard.
`anf_account_name`	String	The Azure NetApp Files account instance identifier.
`subtenant_id`	String	The subtenant ID.
`tags`	–	The key-value pairs.

Retrieve capacity pools

HTTP Method	Path	Description	Path Parameters
`GET`	`/v2.1/anf/pools`	Retrieve capacity pools.	`subtenant_id:` (Mandatory) The subtenant ID to which the ANF account belongs. `tenant_id:` (Optional) Return the capacity pools belonging to the specified tenant. See also Common Pagination parameters.

Required request body parameters: `none`

Request body example:

```
none
```

Retrieve capacity pool by name

Use the method listed in the following table to retrieve a capacity pool by name.

HTTP Method	Path	Description	Parameters
`GET`	`/v2.1/anf/pools/{name}`	Retrieve a capacity pool by name.	`name (string):` (Mandatory) The unique name of the capacity pool. `subtenant_id (string):` (Mandatory) The subtenant ID to which the capacity pool belongs.

Required request body attributes: `none`

Create capacity pools

Use the method listed in the following table to create a new capacity pool.

HTTP Method	Path	Description	Parameters
`POST`	`/v2.1/anf/pools`	Create a capacity pool.	None

Required request body attributes: `name, resource_group, location, size, service_level, anf_account_name, subtenant_id`

Request body example:

```
{
  "name": "string",
  "resource_group": "string",
  "location": "string",
  "size": 10,
  "service_level": "Standard",
  "anf_account_name": "myaccount",
  "subtenant_id": "5d2fb0fb4f47df00015274e3",
  "tags": {
    "key1": "Value 1",
    "key2": "Value 2",
    "key3": "Value 3",
    "keyN": "Value N"
  }
}
```

Modify size of the capacity pool

Use the method listed in the following table to modify the size of the capacity pool.

HTTP Method	Path	Description	Parameters
`PUT`	`/v2.1/anf/pools/{name}`	Modify the size of the capacity pool.	`name (string):` Mandatory: the unique name of the capacity pool.

Required request body attributes: `name, resource_group, location, anf_account_name, size, service_level, subtenant_id`

Request body example:

```
{
  "name": "myaccount",
  "resource_group": "string",
  "location": "string",
  "anf_account_name": "myaccount",
  "size": 4,
  "service_level": "Standard",
  "subtenant_id": "5d2fb0fb4f47df00015274e3",
  "tags": {
    "key1": "Value 1",
    "key2": "Value 2",
    "key3": "Value 3",
    "keyN": "Value N"
  }
}
```

Azure NetApp Files volumes

Azure NetApp Files volume attributes

The following table lists the Azure NetApp Files volume attributes.

Attribute	Type	Description
`id`	String	The unique identifier for the Azure NetApp Files volume.
`name`	String	The name of the Azure NetApp Files volume.
`resource_group`	String	The Azure resource group.
`subtenant_id`	String	The subtenant ID.
`anf_account_name`	String	The Azure NetApp Files account name.
`anf_pool_name`	String	The Azure NetApp Files Pool name.
`location`	String	The Azure location (region/zone).
`file_path`	String	Creation Token or File Path. A unique file path for accessing volume.
`quota_size`	Integer	Maximum storage quota allowed in GiB.

Attribute	Type	Description
`subNetID`	String	The Azure Resource URL for a delegated subnet. Must have the delegation Microsoft NetApp/volumes.
`tags`	–	The key-value pairs.

Retrieve Azure NetApp Files volumes

Use the method listed in the following table to retrieve Azure NetApp Files volumes. Specifying a `tenant_id` returns only the accounts belonging to that tenant.

HTTP Method	Path	Description	Path Parameters
`GET`	`/v2.1/anf/volumes`	Retrieve Azure NetApp Files volumes.	`subtenant_id:` (Mandatory) The subtenant ID to which the ANF volume belongs. `tenant_id:` (Optional) Return the ANF Volumes belonging to the specified tenant. See also Common Pagination parameters.

Required request body parameters: `none`.

Retrieve Azure NetApp Files volume by name

Use the method listed in the following table to retrieve an Azure NetApp Files volume by name.

HTTP Method	Path	Description	Parameters
`GET`	`/v2.1/anf/volumes/{name}`	Retrieve an Azure NetApp Files volume by name.	`name (string):` Mandatory: the unique name of the Azure NetApp Files volume. `subtenant_id:` (string) Mandatory. The subtenant ID to which the Azure NetApp Files volume belongs.

Required request body attributes: `none`

Request body example:

```
none
```

Create Azure NetApp Files volumes

Use the method listed in the following table to create a new Azure NetApp Files volume.

HTTP Method	Path	Description	Parameters
`POST`	`/v2.1/anf/volumes`	Create an Azure NetApp Files volume.	None

Required request body attributes: `name`, `resource_group`, `subtenant_id`, `anf_account_name`, `anf_pool_name`, `virtual_network`, `location`, `file_path`, `quota_size`, `subNetID`

Request body example:

```
{
  "name": "myVolume",
  "resource_group": "string",
  "subtenant_id": "5d2fb0fb4f47df00015274e3",
  "anf_account_name": "myaccount",
  "anf_pool_name": "myaccount",
  "virtual_network": "anf-vnet",
  "location": "string",
  "file_path": "myVolume",
  "quota_size": 100,
  "subNetId": "string",
  "protocol_types": [
    "string"
  ],
  "tags": {
    "key1": "Value 1",
    "key2": "Value 2",
    "key3": "Value 3",
    "keyN": "Value N"
  }
}
```

Managing Cloud Volumes Service for Google Cloud

The `/v2.1/gcp/volumes` API under the Cloud Orchestration category enables you to manage cloud volumes for your Google Cloud instance. Before running this API, ensure that the Cloud Volumes Service account for Google Cloud Platform (GCP) subscription has been enabled for the subtenant.

HTTP Verb	Path	Description	Mandatory parameters/Request body
`GET`	`/v2.1/gcp/volumes`	You can use the GET method to retrieve the details of all the Google Cloud Volumes created for your subtenant's Cloud Volumes Service subscription.	`offset`: The number of items to skip before starting to collect the result set. `limit`: The numbers of items to return. `subtenant_id`: The ID of the subtenant subscribed to Google Cloud. `region`: The region of the subscribed service.
`GET`	`/v2.1/gcp/volumes/{id}`	You can use this method to retrieve the details of a specific Google Cloud Volume created for your subtenant's Cloud Volumes Service subscription.	`id`: The ID of the GCP volume. `subtenant_id`: The ID of the subtenant subscribed to Google Cloud. `region`: The region of the subscribed service.

HTTP Verb	Path	Description	Mandatory parameters/Request body

POST	/v2.1/gcp/volumes	Create a GCP volume for a subtenant. Add the values in the request body to create a volume with the specified parameters.	`
			{
			"subtenant_id": "<ID>",
			"name": "<Volume_name>",
			"region": "<region>",
			"zone": "<zone>",
			"creation_token": "<token>",
			"allowed_clients": "<IP address of the clients allowed to access GCP>",
			"network": "<network details as entered for the GCP subscribed service>",
			"protocol_types": [
			"<Protocol for the connection, such as NFSv3>"
			],
			"quota_gib": <volume quota in bytes>,
			"service_level": "<the type of Performance Service Level, such as standard>",
			"labels": ["<tag_value>"]
			}
			`

HTTP Verb	Path	Description	Mandatory parameters/Request body

PUT	`/v2.1/gcp/volumes/{id}`	Modify a GCP volume already created for a subtenant. Add the volume ID of the volume that you want to modify and the value for the parameters that you want to modify, in the request body.	`
			{
			"subtenant_id": "<ID>",
			"name": "<Volume_name>",
			"region": "<region>",
			"zone": "<zone>",
			"allowed_clients": "<IP address of the clients allowed to access GCP>",
			"quota_gib": <volume quota in bytes>,
			"service_level": "<the type of Performance Service Level, such as standard>",
			"protocol_types": ["<Protocol for the connection, such as NFSv3>"],
			"labels": ["<tag_value>"]
			}
			`

DELETE	`/v2.1/gcp/volumes/{id}`	You can use this method to delete a specific Google Cloud Volume created for your subtenant's Cloud Volumes Service subscription.	`id`: The ID of the GCP volume. `subtenant_id`: The ID of the subtenant subscribed to Cloud Volumes Service for Google Cloud. `region`: The region of the subscribed service.

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